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Zhu Shuyang (朱舒阳)

3D Vision · Geometry Processing · CUDA-Acceleration
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Profile

Graduate student at the University of Tokyo specializing in **3D reconstruction**, **vision**, and **GPU-accelerated computing**. Experienced in **point-cloud and surface processing**, **3D Gaussian Splatting**, and **high-performance computing**, with hands-on development in **C++**, **CUDA**, **Python**, and **PyTorch**.

Publications

An automatic framework for quadrilateral surface reconstruction with partitions from 3D point clouds

Shuyang Zhu, Youlong Li, Jizhao Liu, Bin Li.*
The Visual Computer.
Proposes a four-step framework inspired by polycube parameterization to reconstruct 3D point clouds into quadrilateral meshes with smooth partitions while preserving geometric features.

CTGS: Hybrid Gaussian Splatting for Compact Industrial CT Representation

Shuyang Zhu, Yukie Nagai.*
Accepted by ICPE 2026.
Represents industrial CT volumes with Gaussian Splatting using CT-specific slice supervision, geometric regularization, and CUDA acceleration for volume-based training and analysis.

Selected Projects

Adaptive CT Volume — Ongoing

Adaptive projection-domain CT reconstruction

- Modeled CT volumes as anisotropic octrees for compression and optimized the representation directly from sparse projection data.
- Maintains clear edge features and accurate surface extraction under sparse-projection training; compressed model size is around 8 MB.

CT Simulation Software

High-accuracy and customizable CT simulation

- Built software to simulate CT scans from 3D models with physical accuracy.
- Included lattice generation, defect injection, physics-based compression simulation using FEM, and GPU dynamic batching for large-volume processing.

Porous CT Processor

Pore network modeling + 4D CT analysis

- Designed an analysis and visualization suite with GUI for porous CT datasets.
- Integrated rendering, pore network modeling, 4D tracking of pore evolution, and GPU acceleration using CuPy.

STL Processor

DirectX 12 mesh processing and visualization

- Developed a DirectX 12 application for STL mesh repair and visualization.
- Implemented polycube parameterization, hole filling, normal correction, denoising, and smoothing.

Work Experience

Tencent — Aug 2026 - Present

Game Research & Development Intern, Engine Research

- Developed a new volumetric smoke rendering mode in Unreal Engine 5, enabling efficient smoke volume rendering with global illumination.

Polyphony Digital — Jul 2026

Engineer Intern

- Built a FlashSplat-based 3DGS segmentation tool that extracts object-level segments from already-trained 3D Gaussian Splatting assets within seconds through point selection, without pre-baked segmentation attributes.

Synesthesia — May 2026 - Present

Technical Development Intern

- Delivered a wall-surface inclination metrology workflow that achieves stable millimeter-level accuracy on complex wall point clouds with vegetation, protrusions, and other disturbances.

Fixstars **Apr 2026 - May 2026**
Year-round Intern

- Reduced Open-Sora 2 inference peak VRAM from 39.7 GB to 20.93 GB through lazy loading, unified memory, and cache residency management, enabling efficient execution on 24 GB-class GPUs without quantization.

Education

The University of Tokyo, Tokyo, Japan **Oct 2025 - Present**
M.E. candidate, Nuclear Engineering and Management

Hokkaido University, Sapporo, Japan **Oct 2024 - Mar 2025**
Special Auditor, Department of Electronics and Information Engineering

Lanzhou University, Lanzhou, China **Sep 2021 - Jul 2025**
B.E. in Digital Media Technology

Awards

2025: The University of Tokyo Fellowship
2023: 9th Internet+ Innovation and Entrepreneurship Competition, Gansu Province Gold Award
2022: 13th BlueBridge Cup, Gansu Province Third Prize
2022-2024: Lanzhou University Scholarship